README for Replication Package

Project: U.S. Monetary Policy Spillovers and European Distributional Effects

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Overview

This replication package contains data, MATLAB code, and outputs for the project. The package analyzes the transmission of U.S. monetary policy shocks to euro area downside risk. It constructs predicted GDP-growth quantiles, evaluates in-sample and out-of-sample accuracy across seven quantile-regression specifications, each estimated with and without an autoregressive fitted-quantile component, and produces impulse response functions and counterfactual figures as responses to monetary policy shocks.

- Main data: excel/raw/data_filtered.xlsx, with the baseline panel of 10 euro-area countries, plus monthly and quarterly macro-financial series. The broader excel/raw/data.xlsx file contains the same structure with an 11-country euro-area panel that adds Luxembourg.
- Main derived outputs: excel/processed/pred_q.mat, a time x quantile x country matrix of predicted quantiles used in the figures, and excel/processed/yy.mat, the cumulative GDP-growth outcome used for average-effect comparisons. The current baseline predicted-quantile specification is V4-AR.
- Main scripts: code/quantile_regr/main.m for quantile regressions, accuracy tables, and the downside/upside figure; code/figures/main.m for the main IRFs and counterfactual figures; selected appendix scripts in code/appendix for the U.S. robustness/appendix outputs.
- Existing main outputs: 2 tables in results/tab and 9 PNG figures in results/fig. Existing appendix outputs: 7 PNG figures and pred_q_usa.mat in code/appendix/results_appendix.

Paper title: U.S. Monetary Policy Spillovers and European Distributional Effects

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Repository, DOI, or permanent package link: **TBA**

Data Availability and Provenance Statements

The main input file used by the baseline code is excel/raw/data_filtered.xlsx. The package also includes excel/raw/data.xlsx, a broader euro-area panel with the same sheets and variables, adding Luxembourg to the country panels. The local files contain macro-financial variables in quarterly and monthly frequency.

Statement about Rights

The authors have legitimate access to and permission to use all data used in the manuscript.

The authors have permission to redistribute/publish the data contained in this replication package.

Summary of Availability

All data are publicly available.

Details on Each Data Source

Data name	File / sheet	Observable content	Period	Source / access / license
Baseline workbook	excel/raw/data_filtered.xlsx GDP_panel, CLIFS, other, gov, monthly	10-country euro-area baseline panel; quarterly country data plus monthly/quarterly macro-financial series.	Country panels: 1971Q1-2019Q4. Monthly range: 1971M1-2019M12. Series-specific availability.	Restricted country sample from data.xlsx.
GDP panel	excel/raw/data.xlsx GDP_panel	11 countries: AT, BE, FI, FR, DE, IE, IT, LU, NL, PT, ES. Filtered file excludes LU.	1971Q1-2019Q4	OECD - Quarterly real GDP growth
CLIFS panel	excel/raw/data.xlsx CLIFS	Same 11-country panel as GDP_panel. Filtered file excludes LU.	1971Q1-2019Q4; early missing values for some	ECB DataPortal - CLIFS

			countries.	
Quarterly macro series	excel/raw/data.xlsx other	NFCI, CISS, EONIA, RER, FEDFUNDS, ff4, DEXUSEU, eu_shock, GDPC1, JK, INFO.	1971Q1-2019Q4; series-specific availability.	FRED (NFCI, EONIA, FEDFUNDS, DEXUSEU, GDPC1); ECB (CISS); Miranda-Agrippino and Ricco (2021) (ff4 and INFO); Jarocinski and Karadi (2020) / ECB shock series (JK and eu_shock); Ellahie and Ricco (2017) (RER).
Monthly macro series	excel/raw/data.xlsx monthly	EONIA, ff4, HICP, EU_DOL, IP_EA, FEDFUNDS, JK, Brent, NFCI, CISS, IP_USA, eu_shock, eurostoxx50, ea_bbb_oas_all_fred.	1971M1-2019M12; series-specific availability.	FRED (EONIA, FEDFUNDS, Brent, IP_EA, IP_USA, NFCI, ea_bbb_oas_all_fred); ECB (CISS); Eurostat (HICP); Jarocinski and Karadi (2020) / ECB shock series (JK and eu_shock); Miranda-Agrippino and Ricco (2021) (ff4); market-data source for eurostoxx50 to confirm.
Predicted quantiles	excel/processed/ pred_q.mat	Derived MATLAB output. Baseline specification: V4-AR.	Derived from quarterly data.	Generated by code/quantile_regr/data_build.m.

Dataset List

Data file	Source	Notes	Provided
excel/raw/ data_filtered.xlsx	Constructed from data.xlsx	Main baseline workbook; 10-country sample; sheets: GDP_panel, CLIFS, other, gov, monthly.	Yes
excel/raw/ data.xlsx	Macro-financial sources	Broader workbook; same sheets; 11-country panel includes Luxembourg.	Yes
excel/processed/ pred_q.mat	code/quantile_regr/ data_build.m	Predicted quantiles used by code/figures; regenerated by code/quantile_regr/main.m.	Yes
excel/processed/ yy.mat	code/quantile_regr/ data_build.m	Cumulative GDP-growth outcome for average-effect comparisons.	Yes
code/appendix/results_appendix/ pred_q_usa.mat	code/appendix/ quantile_regr_usa.m	U.S. predicted quantiles for appendix IRF scripts.	Yes

Raw data variables and sheets

- GDP_panel: quarterly real GDP growth-rate panel by country;
- CLIFS: quarterly Country-Level Index of Financial Stress panel by country;
- gov: quarterly government-related country panel included in both raw workbooks;
- NFCI: National Financial Conditions Index;
- CISS: Composite Indicator of Systemic Stress;
- EONIA: Euro OverNight Index Average;
- RER: Real Exchange Rate (Ellahie and Ricco (2017));
- FEDFUNDS: Federal Funds Effective Rate;
- ff4: U.S. monetary policy shock (Miranda-Agrippino & Ricco (2021));
- DEXUSEU and EU_DOL: U.S. dollar/euro exchange-rate series;
- HICP: Harmonized Indices of Consumer Price Index;
- IP_EA and IP_USA: industrial production series for the euro area and the United States;
- GDPC1: U.S. real GDP;
- JK, eu_shock, and INFO: monetary-policy and information-shock series used in the IRF and appendix workflows; Brent: global price of Brent crude; eurostoxx50: Euro Stoxx 50 index; ea_bbb_oas_all_fred: euro-area BBB option-adjusted spread series.

Computational Requirements

Software Requirements

- MATLAB: version used R2025b.
- Required MATLAB toolboxes: Econometrics Toolbox.
- Operating system used for the last run: macOS.
- Included external dependencies: functions in code/routines and code/routines/subroutines. Some subroutines refer to the Panel Data Toolbox. The appendix scripts write to code/appendix/results_appendix.

Runtime Requirements

- Approximate time needed to regenerate all outputs: 45 minutes.

Description of Programs / Code

Path	Purpose	Main outputs / notes
code/quantile_regr/main.m	Quantile-regression entry point.	Runs data_build, table1, table2, figure1.
code/quantile_regr/data_build.m	Builds 0.05:0.05:0.95 predicted quantiles for V1-V7 and AR variants.	Saves pred_q.mat and yy.mat; baseline V4-AR.
code/quantile_regr/table1.m	In-sample accuracy.	In_Sample_accuracy.txt.
code/quantile_regr/table2.m	Out-of-sample recursive accuracy.	OOS_accuracy.txt.
code/quantile_regr/figure1.m	Downside/upside uncertainty.	down_up_ALL.png.
code/figures/main.m	Main IRF/counterfactual entry point.	Runs figure1-figure5; figure7 is separate.
code/figures/figure1.m	EONIA IRF to U.S. MP shock.	eonia_irf.png.
code/figures/figure2.m	Downside-risk IRF to U.S. MP shock; average-effect comparison.	linear_downside_risk_irf.png; comparison_downside_risk_vs_avg_irf.png.
code/figures/figure3.m	No ECB monetary-policy response counterfactual.	counter_no_MP.png; run through main.m.
code/figures/figure4.m	No financial/CISS response counterfactual.	counter_no_CISS.png; run through main.m.
code/figures/figure5.m	Downside-risk IRF to ECB MP shock; average-effect comparison.	linear_downside_risk_irf_ECB.png; comparison_downside_risk_vs_avg_irf_ECB.png.
code/figures/figure6.m	KBO decomposition with CLIFS.	Not run by main.m; no saved output detected.
code/figures/figure7.m	FED-versus-ECB comparison.	comparison_downside_risk_FED_vs_ECB.png; run after main.m.
code/appendix/quantile_regr_usa.m	U.S. predicted quantiles and downside/upside uncertainty.	pred_q_usa.mat; down_up_USA.png.
code/appendix/figureA11_usa.m	U.S. downside-risk IRF to U.S. MP shock.	linear_downside_risk_irf_USA.png; comparison_downside_risk_vs_avg_irf_USA.png.
code/appendix/figureA1_usa.m	U.S. downside-risk IRF to ECB MP shock.	linear_downside_risk_irf_USA_ECB.png; comparison_downside_risk_vs_avg_irf_USA_ECB.png.
code/appendix/figureA2_info.m	Downside-risk IRF to information shock.	linear_downside_risk_irf_INFO.png; comparison_downside_risk_vs_avg_irf_INFO.png.
code/appendix/figureA3.m	Appendix U.S. FED-versus-ECB comparison.	comparison_downside_risk_US_FED_vs_ECB.png; run after relevant U.S. IRFs.

Instructions to Replicators

1. Open MATLAB and set the working directory to the project folder.
2. In MATLAB, move to code/quantile_regr and run main.m. This regenerates pred_q.mat, yy.mat, the two accuracy tables, and down_up_ALL.png.
3. In MATLAB, move to code/figures and run main.m. This regenerates eonia_irf.png, linear_downside_risk_irf.png, comparison_downside_risk_vs_avg_irf.png, linear_downside_risk_irf_ECB.png, comparison_downside_risk_vs_avg_irf_ECB.png, counter_no_MP.png, and counter_no_CISS.png. The counterfactual scripts should be run through main.m because they depend on variables created earlier in the

figure workflow. To regenerate comparison_downside_risk_FED_vs_ECB.png, run figure7.m after main.m in the same MATLAB session.

4. For appendix outputs, move to code/appendix. Run quantile_regr_usa.m first to regenerate pred_q_usa.mat and down_up_USA.png. Then run figureA11_usa.m and figureA1_usa.m for the U.S. shock and ECB shock appendix figures, figureA2_info.m for information-shock figures, and figureA3.m after the relevant U.S. IRF scripts if the comparison variables are needed in the same MATLAB workspace. Compare regenerated outputs with results/fig, results/tab, and code/appendix/results_appendix.

Expected Outputs

Output file	Generated by	Description
results/tab/ In_Sample_accuracy.txt	code/quantile_regr/table1.m	In-sample RMSE, MAE, QS, qwCRPS relative to V4-AR.
results/tab/ OOS_accuracy.txt	code/quantile_regr/table2.m	Out-of-sample RMSE, MAE, QS, qwCRPS relative to V4-AR.
results/fig/ down_up_ALL.png	code/quantile_regr/figure1.m	Downside/upside uncertainty by country.
results/fig/ eonia_irf.png	code/figures/figure1.m	EONIA IRF to U.S. MP shock.
results/fig/ linear_downside_risk_irf.png	code/figures/figure2.m	Downside-risk IRF to U.S. MP shock.
results/fig/ comparison_downside_risk_vs_avg_irf.png	code/figures/figure2.m	Downside risk versus average GDP response, U.S. MP shock.
results/fig/ linear_downside_risk_irf_ECB.png	code/figures/figure5.m	Downside-risk IRF to ECB MP shock.
results/fig/ comparison_downside_risk_vs_avg_irf_ECB.png	code/figures/figure5.m	Downside risk versus average GDP response, ECB MP shock.
results/fig/ counter_no_MP.png	code/figures/figure3.m	No ECB monetary-policy response counterfactual.
results/fig/ counter_no_CISS.png	code/figures/figure4.m	No financial/CISS response counterfactual.
results/fig/ comparison_downside_risk_FED_vs_ECB.png	code/figures/figure7.m	FED-versus-ECB downside-risk comparison; run separately.
code/appendix/results_appendix/ pred_q_usa.mat	code/appendix/quantile_regr_usa.m	U.S. predicted quantiles.
code/appendix/results_appendix/ down_up_USA.png	code/appendix/quantile_regr_usa.m	U.S. downside/upside uncertainty.
code/appendix/results_appendix/ linear_downside_risk_irf_USA.png	code/appendix/figureA11_usa.m	U.S. downside-risk IRF to U.S. MP shock.
code/appendix/results_appendix/ comparison_downside_risk_vs_avg_irf_USA.png	code/appendix/figureA11_usa.m	U.S. downside risk versus average GDP, U.S. MP shock.
code/appendix/results_appendix/ linear_downside_risk_irf_USA_ECB.png	code/appendix/figureA1_usa.m	U.S. downside-risk IRF to ECB MP shock.
code/appendix/results_appendix/ comparison_downside_risk_vs_avg_irf_USA_ECB.p ng	code/appendix/figureA1_usa.m	U.S. downside risk versus average GDP, ECB MP shock.
code/appendix/results_appendix/ linear_downside_risk_irf_INFO.png	code/appendix/figureA2_info.m	Downside-risk IRF to information shock.
code/appendix/results_appendix/ comparison_downside_risk_vs_avg_irf_INFO.png	code/appendix/figureA2_info.m	Downside risk versus average GDP, information shock.
code/appendix/results_appendix/ comparison_downside_risk_US_FED_vs_ECB.png	code/appendix/figureA3.m	Appendix U.S. FED-versus-ECB comparison.

List of Tables and Figures

Paper item	Program	Output file	Note
Table 1	code/quantile_regr/table1.m	results/tab/ In_Sample_accuracy.txt	Numbering to confirm.

Table 2	code/quantile_regr/table2.m	results/tab/ OOS_accuracy.txt	Numbering to confirm.
Figure 1Q	code/quantile_regr/figure1.m	results/fig/ down_up_ALL.png	Baseline downside/upside uncertainty.
Figure 1	code/figures/figure1.m	results/fig/ eonia_irf.png	EONIA IRF.
Figure 2	code/figures/figure2.m	results/fig/ linear_downside_risk_irf.png	U.S. MP shock downside-risk IRF.
Figure 3	code/figures/figure2.m	results/fig/ comparison_downside_risk_vs_avg_irf.png	Downside risk versus average GDP.
Figure 4	code/figures/figure5.m	results/fig/ linear_downside_risk_irf_ECB.png	ECB MP shock downside-risk IRF.
Figure 5	code/figures/figure5.m	results/fig/ comparison_downside_risk_vs_avg_irf_ECB.png	Downside risk versus average GDP.
Figure 6	code/figures/figure3.m	results/fig/ counter_no_MP.png	No ECB MP response.
Figure 7	code/figures/figure4.m	results/fig/ counter_no_CISS.png	No CISS response.
Figure 8 / appendix item to confirm	code/figures/figure7.m	results/fig/ comparison_downside_risk_FED_vs_ECB.png	Numbering to confirm.
Appendix figure to confirm	code/appendix/quantile_regr_usa.m	code/appendix/results_appendix/ down_up_USA.png	U.S. downside/upside uncertainty.
Appendix figure to confirm	code/appendix/figureA11_usa.m	code/appendix/results_appendix/ linear_downside_risk_irf_USA.png	U.S. MP shock IRF.
Appendix figure to confirm	code/appendix/figureA11_usa.m	code/appendix/results_appendix/ comparison_downside_risk_vs_avg_irf_USA.png	U.S. MP comparison.
Appendix figure to confirm	code/appendix/figureA1_usa.m	code/appendix/results_appendix/ linear_downside_risk_irf_USA_ECB.png	ECB MP shock IRF.
Appendix figure to confirm	code/appendix/figureA1_usa.m	code/appendix/results_appendix/ comparison_downside_risk_vs_avg_irf_USA_ECB.png	ECB MP comparison.
Appendix figure to confirm	code/appendix/figureA2_info.m	code/appendix/results_appendix/ linear_downside_risk_irf_INFO.png	Information-shock IRF.
Appendix figure to confirm	code/appendix/figureA2_info.m	code/appendix/results_appendix/ comparison_downside_risk_vs_avg_irf_INFO.png	Information-shock comparison.
Appendix figure to confirm	code/appendix/figureA3.m	code/appendix/results_appendix/ comparison_downside_risk_US_FED_vs_ECB.png	U.S. FED-versus-ECB comparison.

References

- Álvarez, Inmaculada C.; Barbero, Javier and Zofío, José L, (2017). **A Panel Data Toolbox for MATLAB**. *Journal of Statistical Software*. Volume 76, Issue 6, pp 1-27. http://dx.doi.org/10.18637/jss.v076.i06
- Jarociński, M., & Karadi, P. (2020). Deconstructing Monetary Policy Surprises: The Role of Information Shocks. American Economic Journal: Macroeconomics, 12(2), 1–43. https://doi.org/10.1257/mac.20180090
- Miranda-Agrippino, S., & Ricco, G. (2021). The Transmission of Monetary Policy Shocks. American Economic Journal: Macroeconomics, 13(3), 74-107. https://doi.org/10.1257/mac.20180124
- Ellahie, A., & Ricco, G. (2017). Government purchases reloaded: Informational insufficiency and heterogeneity in fiscal VARs. Journal of Monetary Economics, 90, 13-27. https://doi.org/10.1016/j.jmoneco.2017.06.002